Patrick Walukagga

Software & AI Engineer · Backend and AI engineer who takes LLM systems from prototype to production on AWS.

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SUMMARY

Backend software engineer with 9+ years of experience designing and operating scalable backend systems, cloud-native services, and ML-enabled products. Current emphasis on AI and LLM engineering and production inference: dependable APIs, latency and cost trade-offs, observability, and disciplined release practices around models and artifacts. Primary stack is Python, Django, FastAPI, PostgreSQL, Docker, and AWS, with infrastructure as code (Terraform, CloudFormation, Ansible) and GPU inference platforms (RunPod, Modal, Vast.ai, Hugging Face).

TECHNICAL SKILLS

Backend: Python, FastAPI, Django & Django REST Framework, Flask, GraphQL, REST API design, PostgreSQL, Redis & Celery, Testing (pytest)

Data: SQL, Pandas & NumPy, ETL pipelines, Prefect & Airflow, MLflow experiment tracking, DynamoDB single-table design, MySQL

Cloud & Infra: AWS (Lambda, API Gateway, DynamoDB, Cognito, S3, CloudFront, SAM), GCP Cloud Run, Terraform, CloudFormation & Ansible, Docker & Kubernetes, GitHub Actions CI/CD, Nginx & Gunicorn, Bash

AI / ML: LLM-backed services (prompting, tools, structured output), AWS Bedrock & OpenAI-compatible APIs, Model Context Protocol (MCP) servers, Production inference on GPU platforms (RunPod, Modal, Vast.ai, Hugging Face), ASR / TTS backends, RAG & evaluation harnesses, scikit-learn & MLOps tooling

EXPERIENCE

Software Engineer
Apr 2024 – Present

Sunbird AI · Kampala, Uganda

- Build and operate the production APIs behind Sunbird AI's language models: OpenAI-compatible LLM endpoints, speech (ASR/TTS) backends, and an MCP server, deployed on AWS and GCP Cloud Run.
- Run inference across GPU platforms (RunPod, Modal, Vast.ai, Hugging Face Inference Endpoints) with model versioning, warm-pool capacity planning, and cost and latency trade-offs.
- Automate infrastructure and delivery with Terraform, Docker, and GitHub Actions, and add authentication, rate limiting, and usage billing to public APIs.

Python · FastAPI · AWS · GCP Cloud Run · Terraform · Docker · Hugging Face · LLMs

Python/Django Backend Engineer
May 2023 – Jun 2024

Audersity · Kampala, Uganda

- Designed and shipped backend features across Django-based systems with cross-functional product teams.
- Improved API reliability and performance through clean architecture, structured logging, and test coverage.

Python · Django · PostgreSQL · Docker · CI/CD

Data Engineer Graduate Trainee
Feb 2023 – Mar 2024

Cecure Intelligence Limited · Lagos, Nigeria (remote)

- Built ETL pipelines and automated data ingestion workflows feeding analytics and reporting.
- Supported engineering teams delivering data-driven applications.

Python · SQL · Pandas · ETL

Python/Django Developer (contract, MTN Rwanda) Jun 2021 – Jun 2022
HAMWE EA · Kampala, Uganda

- Developed and deployed the Django REST API powering cooperative and farmer onboarding for MTN Rwanda's agriculture digitization platform.
- Designed PostgreSQL schemas for the user, cooperative, collections, and shipment modules and built a USSD module for farmer collection and loan management.
- Ran CI/CD with GitHub Actions to DigitalOcean using Docker, Nginx, and Gunicorn; cleaned field data with Pandas for ingestion.

Python · Django · PostgreSQL · USSD · Docker · GitHub Actions · Pandas

Python/Django Engineer Sep 2021 – Jan 2022
BoldGains · Dubai, UAE (remote)

- Built backend modules for the Blockbrite payment gateway, including the bank transaction integration layer.
- Wrote unit and integration tests for the user and M-Pesa transaction modules to improve reliability.

Python · Django · PostgreSQL · Payments

Software Developer Nov 2017 – Dec 2021
Andela · Kampala, Uganda

- Placed with distributed engineering teams at partner companies building scalable applications.
- Contributed to backend development, testing, and system debugging across several client engagements.

Python · Django · Flask · PostgreSQL · JavaScript

Python/Django Developer May 2020 – Dec 2020
NITA-U IT · Kampala, Uganda

- Built and maintained RESTful APIs for the national IT certification extranet used by providers, institutions, and the certification office.
- Reworked invoice generation across provider types and integrated SMS notifications into the certification workflow.

Python · Django · PostgreSQL · REST APIs

Software Engineer Mar 2019 – Feb 2020
Tarana Wireless, Inc. · Santa Clara, CA (remote)

- Added features to the internal inventory application used to reserve hardware during testing.
- Automated laboratory test processes that validated device metrics before release to residential sites.

Python · Flask · PostgreSQL · Test automation

Junior Software Developer Oct 2018 – Jan 2019
Chartmetric · United States (remote)

- Scraped music analytics data from third-party APIs into PostgreSQL pipelines.
- Wrote SQL exports for client data requests and fixed backend bugs.

Python · PostgreSQL · SQL · Scrapy

EDUCATION

Nkumba University
Bachelor's Degree in Information Technology · 2012 – 2015

CERTIFICATIONS

- GitHub Copilot · GitHub (2025)
- LFS250: Kubernetes and Cloud Native Essentials · The Linux Foundation (2025)
- GenAI Bootcamp (Gold Squad) · ExamPro (2025)
- AWS Certified Solutions Architect – Associate · Amazon Web Services (2024)
- Terraform Beginner Bootcamp (Terraformer) · ExamPro (2023)
- AWS Cloud Project Bootcamp (Teal Squad) · ExamPro (2023)
- AWS Certified Cloud Practitioner · Amazon Web Services (2022)
- PCEP – Certified Entry-Level Python Programmer · Python Institute (2022)

Courses: Retrieval Augmented Generation (RAG), 2026 · LLM Zoomcamp, 2024 · Zero-to-Techie (Data Engineering), 2023 · Applied Data Science Lab, 2023 · MLOps Zoomcamp, 2022 · Applied Data Science I: Scientific Computing & Python (with honors), 2021